

		INSPECTION TEST PLAN FRP- TANKS / PIPES						Document Number :	
								Date: 20-April-2020	
								Rev. 1	
Job:				Equipment:	FRP Tanks / FRP Pipes & Fittings			Manufacturer:	
Client:									
Item/Description	Type of Check	Documents of reference	Quantum of check	Reference documents for acceptance	Records of check	Interventions		Remarks	
						supplier	Suez		
1. Drawings	Fabrication drawing / Design calculation	Configuration drawing / Design Code / Data sheets	100%	SWTS Configuration drawing / Design Code / Data sheets	Appd Dwg	P	R		
2. Raw Material									
a) Resin - (All resins) Verification of characteristics.	Physical/mechanical and chemical analysis	IS6746 (Sp Gvt, Viscosity, Gel Time, Volatile Content, Acid Value)	1 per Batch	Material Specification / Approved drawing	MTC / Lab test reports and heat treatment charts	P	R		
b) Chopped Strand Mat Test	Mechanical	ISO 1887 (Loss of Ignition)	1 per Batch	Material Specification / Approved drawing	MTC / Lab test reports and heat treatment charts	P	R		
	Chemical	ISO 3344 (Moisture Content)							
	Mechanical	ISO 3374 (Mass per unit area)							
c) Woven Roving Material/ Fibre Glass	Mechanical	ISO 3374 (Mass per unit area)	1 per Batch	Material Specification / Approved drawing	MTC / Lab test reports	P	R		
	Chemical	ISO 3344 (Moisture Content)							
d) PP Plates and Pipes (As applicable)	Physical/mechanical and chemical analysis	Approved drawing / Material Specification	1 per Batch	Material Specification / Approved drawing	MTC / Lab test reports	P	R		
3. In process									
a) Mould-Dimn. & Surface Quality	Dimension / orientation	Approved drawing	100%	Approved drawing	Inspection report	P	R		
b) Coupon testing	Mechanical tests - Glass content/tensile strength/Lap shear strength/Barcol hardness	Approved drawing/Manufacturer specification	100%	Approved drawing/Manufacturer specification	Lab test report /Inspection report	P	R		
4. Final Inspection									
a) Visual & dimension	Workmanship color verification/ dimensions / nozzle orientation / Lifting Lugs/ Base plate, Support Cleats	Approved drawing	100%	Approved drawing	Inspection report	P	W*	*For Pipes & Fittings - Random witness shall be done Based on Customer Call during visit.	
b) Water fill test / Hydrotest	Leakage test (Hydro test 1 hr duration)	BS 4994 (Standard/ Approved Drawing)	100%	No leakage	Inspection report	P	W		
c) Acceptance test (Laminate Testing)	Separate Cutout sample to be used for laminate testing	BS 4994 :B6 (Ultimate Tensile Unit Strgt)	Each	≥ 200 N/mm2 width per Kg/m2	Inspection report	P	W/R		
		BS 4994 :B9 (Lap Shear Strgt)	Each	≥ 7 N/mm2	Inspection report	P	W/R		
	Sample should be made during same process of making tank or pipes	ASTM D2584 (Glass Content)	Each	40% (+ - 10%)	Inspection report	P	W/R		
		ASTM D2583 (Barcoal Hardness)	Each	Min 30BHU	Inspection report	P	W/R		
d) Final inspection	Verification of Tag, name plate, workmanship, Marking of individual parts , and document completion	As per approved drawings/manufacturer procedure/Purchase Order, approved ITP	100%	Approved drawings/manufac turer procedure	Visual/Photos/ Quality Dossier	P	W		
e) Painting	1. UV Pigmentation 2. DFT/ Visual	As per approved drawings/Purchase Order	100%	Approved drawings & Specifications	Painting Report	P	W		

e) Packing	Name plate & marking	Approved drawing / Manufacturers/ Suez WTS Procedure	100%	Approved drawing / Manufacturers/ Suez WTS Procedure	Visual, Dimensional, Packing List, Photos	P	R*	1. Photos to be shared before packing with name plate 2. Photos to be shared After packing
Abbreviation:								
P	Perform				R	Review		
W	Witness (Proceed to next staSWTS based on Witness by Suez WTS or concurrence from S WTS after intimation)				RW	Random Witness		